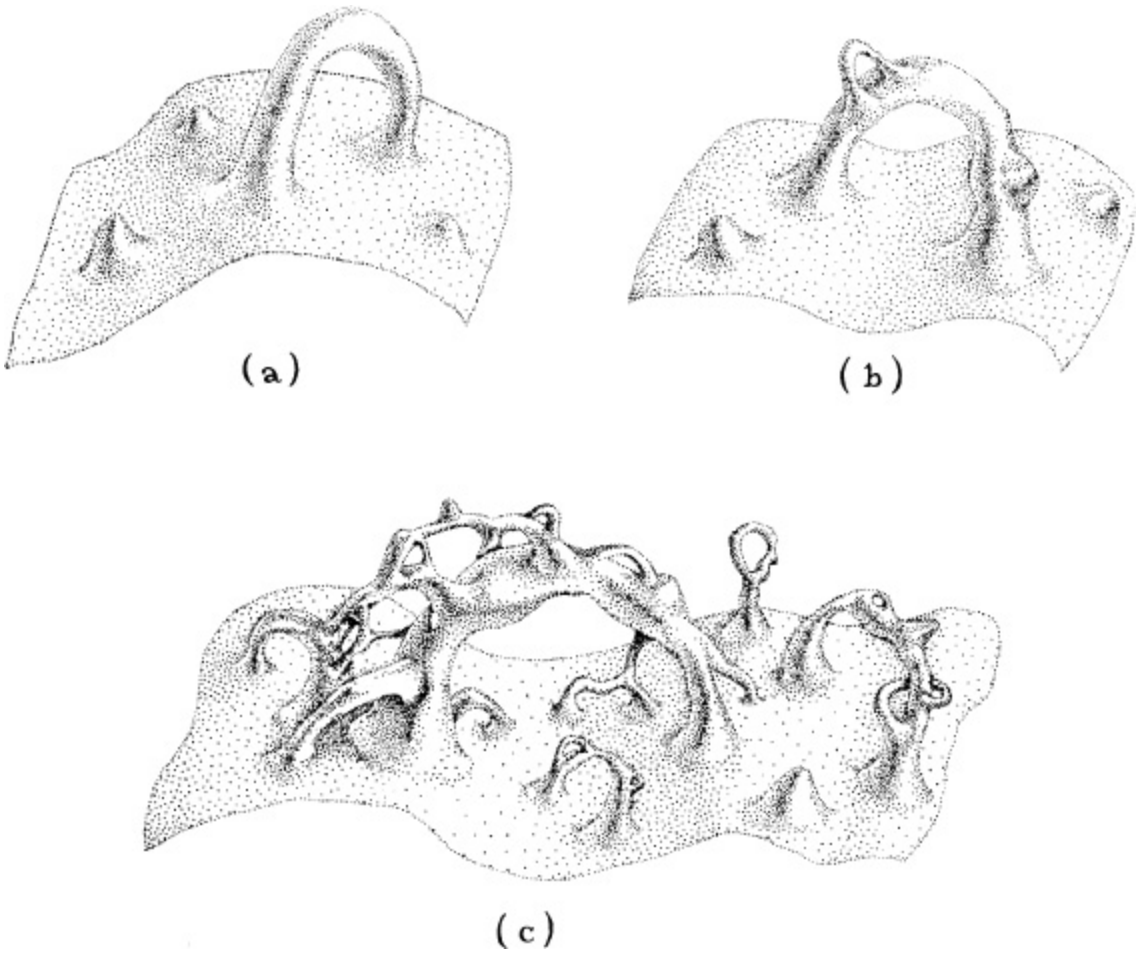
To recapitulate, at the center of a black hole, in the spacetime region where the oscillating BKL tidal forces reach their peak, there resides a singularity: a region in which time no longer exists, and space has given way to quantum foam.



13.7 Embedding diagrams illustrating the quantum foam that is thought to reside in the singularity inside a black hole. The geometry and topology of space are not definite; instead, they are probabilistic. They might have, for example, a 0.1 percent probability for the form shown in (a), a 0.4 percent probability for (b), a 0.02 percent probability for (c), and so on.

One task of the laws of quantum gravity is to govern the probabilities for the various curvatures and topologies within a black hole's singularity. Another, presumably, is to determine the probabilities for the singularity to give birth to "new universes," that is, to give birth to new, classical (non-quantum) regions of spacetime, in the same sense as the big bang singularity gave birth to our Universe some 15 billion years ago.

How probable is it that a black hole's sirlgularity will give birth to "new universes"? We don't know. It might well never happen, or it might be quite